

CENTER (5th Order Ambisonics Vocal Composition)

1. Artistic Concept

CENTER is an immersive vocal composition about love, emotional orientation, and spatial perception. The piece explores the idea of one person becoming the emotional center of the author's world, using 3D audio to physically translate this sensation to the listener.

The lyrics reference positions such as front, behind, left, right, near, and far; turning the voice itself into part of the narrative.

All musical elements are created exclusively with the human voice, including harmonies, vocal bass, percussion, adlibs, and vocal chops.

2. Musical Composition and Workflow

The composition was first produced and recorded in GarageBand using multiple dry vocal layers. The project is based on the progression:

Fmaj7(add9) - Am7 - Dsus2 - Gm7-Fadd9

The tonal center is mainly around F major, using extended jazz-pop harmony to create a soft and atmospheric sound.

Each chord note was individually recorded with the voice and later spatially distributed around the listener in Reaper using the Ambisonics template and the IEM StereoEncoder.

3. Use of Voice as Instrument

The entire composition is built using vocal material. Different vowels are intentionally used as different timbral sources because each vowel contains unique formants and resonant frequencies.

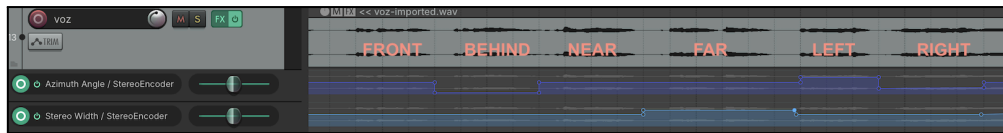
These formants make each vocal layer behave similarly to a different instrument, enriching the harmonic content while maintaining a unified sonic identity.

4. Spatial Design

The spatialization follows a clear hierarchy where not all elements move equally. The objective is to maintain clarity while using movement as a narrative tool.

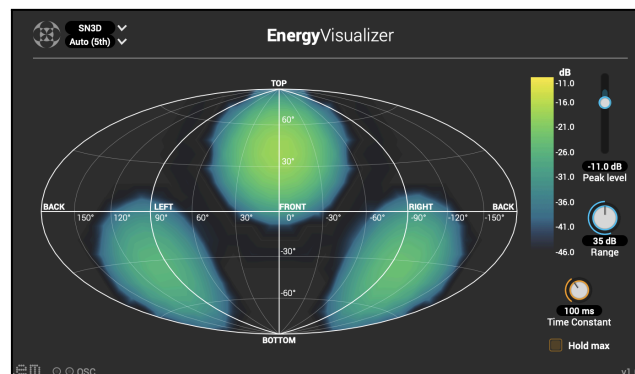
Main Vocal

The lead vocal is the most dynamic spatial element and usually remains focused in front of the listener. Words such as front, behind, left, right, near, and far are reinforced through azimuth movement, width variation, and reverberation. During the chorus, the voice moves more freely around the listener to create a floating sensation.



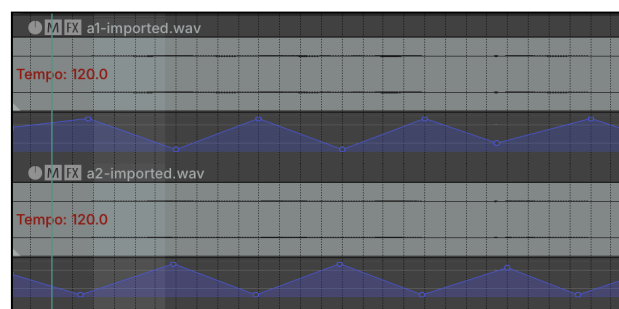
Harmonic Layer

The harmonic layer acts as the immersive foundation of the piece. Each chord note is distributed spatially around the listener, with two notes positioned laterally and slightly below, and one note placed in front with positive elevation, creating openness and depth. These layers remain mostly static to maintain stability.



Vocal chops

Two vocal chops layers are placed at approximately $+45^\circ$ elevation and rotate continuously through opposite azimuth automation, creating subtle counter-movement and rhythmic motion.

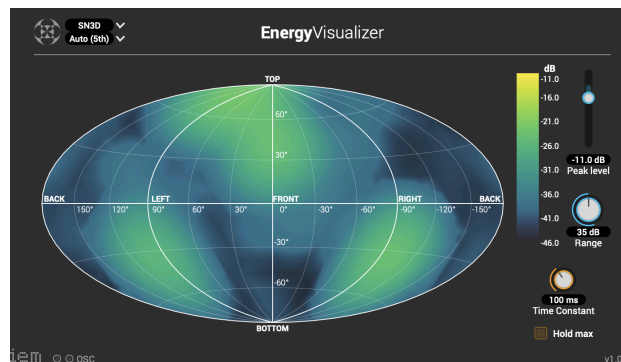


Bass and Percussion

At the middle of the composition, vocal bass and percussion elements are introduced to increase rhythmic energy. The bass and kick remain centered to preserve low-frequency clarity and avoid frequency competition, while the snare and hi-hat vocals are positioned slightly above with subtle widening to enhance the immersive sensation.

5. Reverb and Spatial Depth

Reverb is used selectively to reinforce emotional distance, especially during the word “*far*”, where additional spatial depth is introduced. Instead of using standard stereo reverb directly on the track, an Ambisonic reverb was implemented using the IEM **FdnReverb** plugin to create a large and diffuse immersive space while maintaining the clarity of the lead vocal. The rest of the composition remains relatively dry to preserve focus and localization.



6. Lyrics

The following lyrics summarize the central emotional concept of the composition through repetitive spatial references and simple melodic phrases. For simplicity, only the main lyrical sections are included ([A],[B],[C]).

CENTER

[A] In everything I see,
In every place I go,
You're the one for me,
You're the center of my world.

[B] If I'm in front of you,
If I'm behind,
If I am near you,
If I am far,
Maybe on your left,
Maybe on your right,
Always around.

[C] You are my center,
You are my center,
No matter where I am.
You are my center,
You are my center,
No matter where I've been.

7. Conclusion

CENTER uses spatial audio as part of the emotional and musical narrative rather than as a simple technical effect. Through vocal-only instrumentation, harmonic spatialization, and movement linked to the lyrics, the composition transforms the idea of emotional centrality into a physical listening experience.